On Algebraic Integral For Dummies

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Abstract

We explain the essence of the integral axiom and provide examples of integrating elementary functions using the new integration method.

Introduction

My paper "On Algebraic Integral" [1] has not received the circulation it deserves. Moreover, it appears that guys from the so-called "academic community" — an international governmental organization designed to slow down scientific progress to protect the governments from dissent — these guys have failed to understand my discovery. At the very least, I have received no significant response from them.

That is why I decided to write a simplified version of the paper — a version for dummies — to show the essence of the discovery in the most basic terms. I also want to highlight some of its possible applications. Of course, the applications of that discovery are not limited to the facts presented in the paper. Many of them I cannot even imagine — just as Arthur Cayley could not have imagined the use of matrices in artificial intelligence. I will only point out the most obvious ones.

Moreover, as a researcher, I see my primary mission as exposing the bankruptcy of the academic community (together with the contemporary educational system) and its immeasurable harm to society. I am currently working on a book on this very subject and do not plan to be greatly distracted from that work. I am writing this article only as an explanation to accompany the main paper "On Algebraic Integral" [1] — in response to requests I have received from several people.

What is the integral axiom?

In the first part of the paper, we will show that the integral is not a function of a single variable, but an operation $f \triangleright g = \int f \, dg$. While the method of change of variables is nothing other than the distributivity of composition over that operation:

$$(f \triangleright g) \circ h = (f \circ h) \triangleright (g \circ h)$$

Let's begin.

Let $\langle R, \cdot, \circ \rangle$ be a near-ring. That is, a ring with only right distributivity. That is, R such a structure that

$$\forall f, g, h \in R : (f \cdot g) \circ h = (f \circ h) \cdot (g \circ h)$$

Let $d \colon R \to R$ be a function satisfying the chain rule, that is

$$\forall f, g \in R : d(f \circ g) = \big(d(f) \circ g\big) \cdot d(g)$$

Let $\int \colon R \to R$ be a function opposite to d. Then

Theorem. $\forall f, g, h \in R :$

$$\Big(\int f \cdot d(g) \Big) \circ h = \int (f \circ h) \cdot d(g \circ h)$$

Lemma. $\forall f, g \in R :$

$$\int (f) \circ \int (g) = \int \Big(f \circ \int (g) \cdot g \Big).$$

Proof. For any $x, y \in R$, chain rule gives:

$$d(x \circ y) = d(x) \circ y \cdot d(y).$$

Substitute $x = \int f$, $y = \int g$:

$$d\Big(\int f \circ \int g \Big) = \Big(f \circ \int g \Big) \cdot g.$$

Applying $\int$ to both sides yields

$$\int f \circ \int g = \int \big(f \circ \int g \cdot g\big). \quad \square$$

Proof of Theorem:

$$\Big(\int f \cdot d(g)\Big) \circ h = \Big(\int f \cdot d(g)\Big) \circ \int d(h)$$

Applying Lemma:

$$= \int \Big(\big[f \cdot d(g)\big] \circ \Big(\int d(h)\Big) \cdot d(h)\Big)$$

$$= \int \Big(\big[f \cdot d(g)\big] \circ h \cdot d(h)\Big)$$

By right distributivity in R:

$$= \int \Big(\big[f \circ h\big] \cdot \big[d(g) \circ h\big] \cdot d(h)\Big)$$

And reduce the expression by the chain rule for d:

$$= \int \Big(\big[f \circ h\big] \cdot d(g \circ h)\Big)$$

Or in other words:

$$(f \triangleright g) \circ h = (f \circ h) \triangleright (g \circ h)$$

— We call it "integral axiom". ■

There is also a converse statement: informally speaking, for any distributive operation, there exists a corresponding function that satisfies the chain rule. For details see Theorem 2 in "On Algebraic Integral" [1].

Examples

We will now integrate several elementary functions using the integral axiom, to demonstrate how the theorem simplifies integration by turning it into a purely algebraic process.

Example 1.

$$\frac{1}{x\ln(x)} \triangleright x = \frac{1}{\ln(x)} \triangleright \ln(x)$$

- put $\frac{1}{x}$ under the differential

$$= [\frac{1}{x} \circ \ln(x)] \triangleright \ln(x) = (\frac{1}{x} \triangleright x) \circ \ln(x)$$

- apply the integral axiom.

$$= \ln(\ln(x))$$

- for details about integrating of $\frac{1}{x}$ see Lemma 4.8 in "On Algebraic Integral" [1]

Example 2.

$$x^2(x^3+1)^{\frac{1}{3}} \triangleright x = (x^3+1)^{\frac{1}{3}} \triangleright \frac{1}{3}x^3$$

- put x^2 under the differential

$$= \frac{1}{3}[x^{\frac{1}{3}} \circ (x^3+1) \triangleright (x^3+1)] = \frac{1}{3}[(x^{\frac{1}{3}} \triangleright x) \circ (x^3+1)]$$

- use homogeneity of d and the axiom of integral. Note that composition has priority over the integral.

$$= \frac{1}{3}[(\frac{3}{4}x^{\frac{4}{3}}) \circ (x^3+1)] = \frac{1}{4}(x^3+1)^{\frac{4}{3}}$$

Example 3.

$$\frac{\sin(x)}{1+\cos^2(x)} \triangleright x = \frac{1}{1+\cos^2(x)} \triangleright (-\cos(x))$$

$$= -(\frac{1}{1+x^2} \circ \cos(x) \triangleright \cos(x)) = -(\frac{1}{1+x^2} \triangleright x) \circ \cos(x)$$

- the integral axiom

$$= -\arctan(\cos(x))$$

Simplicity of finding integrals demonstrated in Examples 1–3 suggests that the proposed algebraic method of integration can be turned into a fast and efficient symbolic integration algorithm for computer algebra systems and symbolic libraries such as MATLAB and SymPy. It remains an open question whether the method can accelerate numerical integration — which, in theory, could lead to the optimization and acceleration of neural networks and other mathematical systems.

References

[1] N. Umerov *On Algebraic Integral*. Zenodo, 2025. DOI: 10.5281/zenodo.17376700